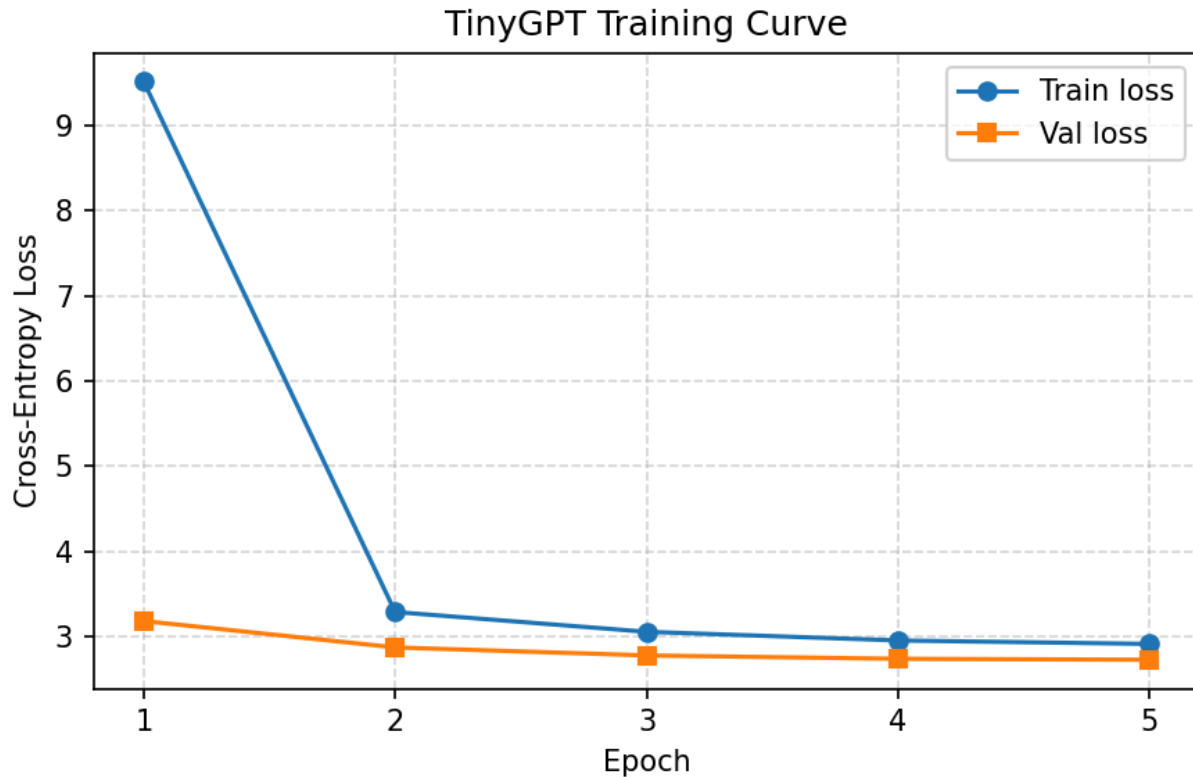


R1. Training Curve



Epoc h	Train Loss	Val Loss
1	9.5202	3.1801
2	3.2860	2.8699
3	3.0533	2.7758
4	2.9515	2.7361
5	2.9110	2.7252

From the training results we can determine that the model is learning normally. Both of the losses decrease after every epoch and the validation loss never increases. The final val loss dropped from 3.1801 at epoch 1 to 2.7252 at epoch 5 (~14% reduction). More epochs will likely steadily drop the loss further.

R2. Generated Text Analysis

Epoch 1 Snippet - ono the iedRs!ccoHhl c lcot yo gH tL t e e t mke

- In epoch 1 we get almost entirely noise, besides the word 'the' none of the other combination of letters are recognizable english, though I do notice there are many more t's and e's than other characters.

Epoch 3 Snippet - LlsKoffg semechet eSd Lal lIdin\lvi:\nThiofin t an C riore t ous athanchalls.
\nttouneve woncy athemyotondi wme youp oving e g t th
\nMfanor BRstinr thoovemeyve an le , tiggangy mr of Move, adere senthidron P diso l and
bisRRlOmh,, o aerest blksee lty ??st r cerothe nd as stdech o asina yesthinresed thi",

- In epoch 3 we get still a lot of noise, but we see less singular letters and words are starting to take form. There are some recognizable english words such as "Move", "an", "and". There is more frequent punctuations but mostly sentences are still incoherent.

Epoch 5 Snippet - Tat he l avinim soreane iwonoXvLteof we ifo whe se,\n
\nMMMWhienr reandoti anivt th woengiaalethammendikpf t p tinn c st lot s ongs s n tor
ensdcnhy ttund,k l, the xnd s sinne and oucumin, berdeomd ill tedhiona. d ,
\nBaeere arorenm cond, l ichades nr ganr at,\nEs ainghsod, yo& aalot w s t asen, mar d de",

- In epoch 5 there are more recognizable words such as "we", "songs", though it is still not that good. Punctuation is falling in more "plausible" locations suggesting that the model may be learning more about sentence structure, but the model still isn't about the put together a coherent sentence, even after epoch 5.

R3 - Temperature Analysis

T = 0.5 snippet - T we thavee e h se withe s the RI wh are on sth the thand me the ce athathes t
the he th te!

- T = 0.5 seems very repetitive. The generation is dominated by the word "the" and the letters "s" and "t". It seems like the model is producing low diversity text.

T = 1.0 snippet - Tat he l avinim soreane iwonoXvLteof we ifo whe se, ...the xnd s sinne and
oucumin, berdeomd ill

- A little more diverse than T = 0.5, there are still coherent words but less than T=0.5 this is likely due to less repetition and trying to make new words.

T = 1.5 snippet - td RnuSndv?'ETingrsz ulHEUUUIhmaofoos rint MaGatim, WatL? waton
erospPdo,

- Highly chaotic, a lot of diversity as opposed to the previous temperatures. There are also a lot of random upper case letters which was not seen for other temperatures

Why does higher temperature lead to higher diversity ?

- At each step the model samples from $P(x) = \text{softmax}(z/t)$. When $t < 1$ the logits are amplified, concentrating probability on the most likely token causing a repetitive output

as seen with $T = 0.5$. When $t > 1$ the logits are compressed making all tokens equally likely making the output more diverse but less coherent, as seen in $t = 1.5$.

R4 - Reflection

From the results TinyGPT was able to learn a number of things. It learned different levels of character n grams, where bi and tri-gram words like “and” “the” “if” started appearing consistently. The model also learned somewhat where to put spaces and punctuation. Though a lot of the generation isn’t coherent the spacing and punctuation follows an english like sentence structure. Also by epoch 5 the model started capitalizing words after newlines, learning that letters should be capitalized when they are at the start of a sentence. A limitation of the model is that it cannot form grammatically correct sentences or maintain coherence across more than a few characters. Operating at the character level means there is no direct access to word-level semantics, and every word must be reconstructed character by character. The 128-character context window also limits the ability to capture long-range structure like multi-turn dialogue.

How a larger model could help: More layers and wider embeddings would give more capacity for complex grammatical patterns. Most significantly, subword tokenization would treat common words as single tokens, reducing effective sequence length and allowing the attention mechanism to operate at the word and sentence level within the same context window.